

BLUESTOCK FINTECH

MUTUAL FUND ANALYTICS PLATFORM

Final Project Report

End-to-end data engineering, ETL, exploratory data analysis, fund performance and risk analytics, advanced investor analytics, and Power BI dashboarding.

Prepared from the Bluestock capstone brief, supplied project datasets, executed notebooks, and Power BI screenshots.

GitHub status: the project was pushed to origin/main with commit d9526ba after the Power BI dashboard and advanced analytics work were committed.

Academic / educational use only. This report is not investment advice.

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1. Executive Summary

The Bluestock Mutual Fund Analytics Platform is an end-to-end fintech analytics solution that converts fragmented mutual-fund data into an integrated analytical workflow. The project uses 40 mutual-fund schemes and ten datasets covering fund master data, daily NAV history, fund-house AUM, monthly SIP inflows, category inflows, industry folios, scheme performance, investor transactions, portfolio holdings and benchmark indices.

The capstone brief defines eight primary objectives: build an ETL pipeline, design a normalized SQL schema, perform EDA, compute performance and risk metrics, build an interactive BI dashboard, analyse investor transaction patterns, compare fund returns with benchmarks, and document the full solution. The requested architecture follows Extract -> Transform -> Load -> Analyse -> Visualise, with SQLite as the development database and Power BI as the dashboard layer.

The completed analytics combine return and risk measures rather than relying on headline performance alone. Daily returns are derived from NAV, CAGR is evaluated over multiple horizons, Sharpe and Sortino measure risk-adjusted performance, Alpha and Beta compare funds against NIFTY 100, and Maximum Drawdown and Tracking Error add downside and benchmark-relative context. Advanced analytics extend the framework to Historical 95% VaR/CVaR, 90-day Rolling Sharpe, investor cohorts, SIP continuity, risk-appetite recommendations and HHI concentration analysis.

Project Snapshot	Value
NAV observations	46,000
Funds	40
Date range	2022-01-03 to 2026-05-29
Investor transactions	32,778
SIP transactions	19,716
Unique SIP investors	4,762
6+ SIP eligible investors	1,362
Project-defined at-risk investors	1,332
At-risk rate	97.80%

Evidence note: the supplied screenshots visibly show successful dataset loading, daily-return validation, CAGR comparison, benchmark/tracking-error analysis, VaR/CVaR, rolling Sharpe, cohort analysis and SIP continuity. The four Power BI pages are also supplied as rendered screenshots and are reproduced in Appendix A.

2. Introduction and Business Context

Mutual fund analytics sits at the intersection of financial data engineering, statistics and business intelligence. A fund can be evaluated from several viewpoints: the evolution of its NAV, its return across a meaningful investment horizon, the volatility associated with that return, its downside loss characteristics, its behaviour relative to a benchmark, and its portfolio concentration. A dashboard that exposes only one of these views can lead to incomplete decisions.

The Bluestock capstone frames the business context around the rapid growth of the Indian mutual-fund industry and the need to democratise investment analytics for retail and institutional users. The brief highlights fragmented data, the performance-comparison gap, the lack of benchmark tracking, investor-behaviour blind spots and slow static reporting as the core problems to solve.

2.1 Business Questions Addressed

- How has NAV evolved across the selected fund universe?
- Which funds show stronger multi-year growth and risk-adjusted performance?

- How much do fund returns deviate from NIFTY 50/NIFTY 100 benchmarks?
- Which schemes exhibit greater downside-tail risk?
- What patterns exist across SIP cohorts, demographics and geographies?
- Which investors show weaker SIP continuity under the project rule?
- Which equity portfolios exhibit greater sector concentration?
- How can the same data be exposed through an interactive executive dashboard?

3. Problem Statement and Objectives

The capstone identifies five practical problems. P1 is data fragmentation: NAV, AUM, SIP flows and holdings are available in different formats and locations. P2 is the performance comparison gap: raw NAV requires transformation before funds can be compared on a risk-adjusted basis. P3 is the benchmark-tracking gap: fund and index histories must be joined to evaluate active performance. P4 is the investor behaviour blind spot: demographic and geographic patterns are difficult to observe from isolated transactions. P5 is slow reporting: static reports do not provide self-service drill-down.

3.1 Objectives

Objective	Outcome
O1	Automate ingestion and ETL from raw AMFI/project data.
O2	Design a normalized five-table star schema for mutual-fund data.
O3	Perform comprehensive EDA and document key findings.
O4	Compute fund performance and risk metrics.
O5	Build a four-page interactive Power BI dashboard.
O6	Analyse investor transaction patterns and demographics.
O7	Benchmark funds and calculate Alpha/tracking error.
O8	Produce the final PDF report and 12-slide presentation.

4. Mutual Fund Concepts and Theoretical Foundation

4.1 Net Asset Value (NAV)

NAV is the per-unit value of a mutual-fund scheme. In conceptual terms, it is driven by the net value of the scheme assets after liabilities divided by the units outstanding. In this project, daily NAV is the primary time series from which return and risk measures are derived.

4.2 Assets Under Management (AUM)

AUM represents the assets managed by a fund house or scheme. The project includes AUM in crore and lakh-crore units. Unit labels matter because an industry-level AUM headline and a scheme-level AUM figure represent different scales.

4.3 Systematic Investment Plan (SIP)

SIP is a recurring investment mechanism. From an analytics perspective it generates a transaction history that can be segmented by time, demographic group, geography and investor continuity.

4.4 Benchmark

A benchmark is a market reference used to judge relative performance. NIFTY 50 and NIFTY 100 are used in the project to contextualise fund returns.

4.5 Daily Return

Daily return converts the NAV level into a percentage return series suitable for statistical analysis:

$$\text{Daily Return} = (\text{NAV}_t / \text{NAV}_{t-1}) - 1$$

4.6 CAGR

CAGR expresses a multi-period growth rate as a constant annualised rate:

$$\text{CAGR} = (\text{NAV}_{\text{end}} / \text{NAV}_{\text{start}})^{(1/n)} - 1$$

4.7 Sharpe Ratio

Sharpe measures excess return per unit of total volatility. The project brief specifies a 6.5% risk-free-rate proxy and annualisation by $\sqrt{252}$:

$$\text{Sharpe} = (R_p - R_f) / \text{Std}(R_p)$$

4.8 Sortino Ratio

Sortino replaces total volatility with downside deviation, so it emphasises harmful return variability.

$$\text{Sortino} = (R_p - R_f) / \text{Downside Std}$$

4.9 Alpha and Beta

The project models fund returns against NIFTY 100 using OLS regression. Alpha is the annualised intercept and Beta is the slope. Beta describes benchmark sensitivity; Alpha captures the modeled excess component after accounting for benchmark movement.

$$R_{\text{fund}} = \text{Alpha} + \text{Beta} * R_{\text{benchmark}} + \text{error}$$

4.10 Maximum Drawdown

Maximum Drawdown measures the largest peak-to-trough decline in the NAV path:

$$\text{Maximum Drawdown} = \min(\text{NAV} / \text{Running Maximum} - 1)$$

4.11 Tracking Error

Tracking Error measures the variability of active returns, where active return is fund return minus benchmark return. It is a useful consistency measure when the objective is benchmark-relative investing.

$$\text{Tracking Error} = \text{Std}(\text{Return}_{\text{fund}} - \text{Return}_{\text{benchmark}})$$

4.12 Historical VaR and CVaR

Historical Value at Risk uses the empirical return distribution. At 95% confidence, VaR is the 5th percentile of daily returns. CVaR is the mean return among observations at or below the VaR threshold. These measures focus on the left tail rather than overall volatility.

$$\text{VaR}_{95} = \text{5th percentile of daily returns}$$

$$\text{CVaR}_{95} = \text{mean}(\text{Return} \leq \text{VaR}_{95})$$

4.13 Rolling 90-Day Sharpe

A rolling Sharpe recalculates risk-adjusted performance over a moving window. The project uses a 90-day window and annualises by $\sqrt{252}$:

$$\text{Rolling Sharpe} = \text{rolling_mean} / \text{rolling_std} * \sqrt{252}$$

4.14 Investor Cohorts

Cohort analysis groups investors by a shared first-transaction year. This allows 2024 and 2025 investors to be compared on total investment, average SIP amount and fund preference.

4.15 SIP Continuity

For investors with at least six SIP transactions, the project computes the average gap between consecutive SIP dates. An average gap greater than 35 days is flagged as at-risk. This is an analytical monitoring rule defined by the capstone, not a validated prediction model.

4.16 Herfindahl-Hirschman Index (HHI)

HHI measures portfolio concentration by summing squared sector weights. Squaring gives greater influence to large positions, so a higher HHI indicates greater concentration:

$$\text{HHI} = \text{Sum}(\text{weight}_i^2)$$

5. Data Sources and Dataset Inventory

The capstone lists AMFI India, mfapi.in, mfddata.in, NSE India, BSE and AMFI monthly notes as the main public sources. The investor transaction dataset is simulated for educational analysis, while fund metadata and market data are based on public sources.

Dataset	Rows	Purpose
01_fund_master.csv	40	40 real schemes; AMC, category, plan, benchmark, expense ratio and risk
02_nav_history.csv	46000	Daily NAV for 40 schemes, Jan 2022-May 2026
03_aum_by_fund_house.csv	90	Quarterly AUM for 10 fund houses
04_monthly_sip_inflows.csv	48	Monthly SIP inflow and account measures
05_category_inflows.csv	144	Category-wise net inflows
06_industry_folio_count.csv	21	Industry folio milestones
07_scheme_performance.csv	40	1/3/5-year returns and risk metrics
08_investor_transactions.csv	32778	SIP, lumpsum and redemption transactions
09_portfolio_holdings.csv	322	Equity holdings, weights and sectors
10_benchmark_indices.csv	8050	Daily benchmark index close values

The uploaded capstone brief states that 02_nav_history.csv is anchored to real historical values from mfapi.in, 03_aum_by_fund_house.csv is based on AMFI quarterly reports, 04_monthly_sip_inflows.csv comes from AMFI monthly notes, and 08_investor_transactions.csv is simulated for 5,000 investors. This distinction is carried into the interpretation of results.

6. System Architecture and ETL Pipeline

The project uses the standard Extract -> Transform -> Load -> Analyse -> Visualise pattern. This design separates acquisition from transformation, storage from analytics, and computation from reporting. It also supports repeatability: the same processed data can feed SQL, Python notebooks and Power BI.

End-to-End Mutual Fund Analytics Architecture

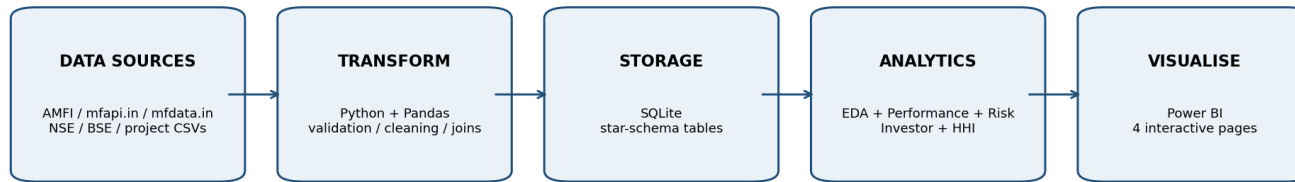


Figure 1. End-to-end architecture specified by the capstone and implemented in the project.

6.1 Extract

Extraction covers the supplied project CSVs plus public/API sources such as mfapi.in. The Day 1 task requires loading all ten CSVs, inspecting shape/dtypes/head, fetching live NAV for selected schemes and validating AMFI codes against the master list.

6.2 Transform

Transformation includes date parsing, numeric validation, duplicate checks, standardising transaction types, validating NAV positivity, joining fund metadata, and deriving daily returns. The brief also specifies forward-filling missing NAV caused by weekends/holidays after creating the full date range.

6.3 Load

Cleaned datasets are loaded into SQLite using the required star-schema-style tables. The database is indexed by amfi_code and date so that analytical queries remain efficient.

6.4 Quality Checks

Data-quality check	Observed result
Check	Result
NAV rows	46,000
Missing NAV values	40
Duplicate NAV rows	0
Duplicate fund-date records	0
Unique funds	40

7. Database Design and SQL Analytics

The database provides a controlled relational layer between source files and analytics. The capstone describes a five-table core star schema with a fund dimension, date dimension, NAV fact, transaction fact and performance fact, plus supporting AUM, SIP and portfolio fact tables.

Table	Type	Key fields	Purpose
dim_fund	Dimension	amfi_code	Scheme master and classifications
dim_date	Dimension	date_id	Calendar attributes for date

			analysis
fact_nav	Fact	amfi_code + date	Daily NAV and derived returns
fact_transactions	Fact	tx_id	Investor transaction events
fact_performance	Fact	amfi_code + as_of_date	Precomputed performance/risk
fact_portfolio	Fact	amfi_code + stock + date	Holdings and sector weights
fact_aum	Fact	fund_house + date	Fund-house AUM history
fact_sip_industry	Fact	month	Industry SIP measures

7.1 Example Analytical SQL

The capstone requires ten analytical SQL queries. Representative queries are shown below.

```
SELECT fund_house, SUM(aum_crore) AS total_aum FROM fact_aum GROUP BY fund_house ORDER BY total_aum DESC LIMIT 5;
```

```
SELECT strftime('%Y-%m', date) AS month, AVG(nav) AS avg_nav FROM fact_nav GROUP BY month ORDER BY month;
```

```
SELECT state, SUM(amount_inr) AS total_transaction_amount FROM fact_transactions GROUP BY state ORDER BY total_transaction_amount DESC;
```

```
SELECT scheme_name, expense_ratio_pct FROM dim_fund WHERE expense_ratio_pct < 1.0 ORDER BY expense_ratio_pct;
```

8. Exploratory Data Analysis

EDA is used to understand the shape of the data before applying more advanced models. The capstone requires at least 15 publication-quality charts spanning NAV, AUM, SIP, investor, geographic, folio, correlation and sector views.

8.1 NAV Behaviour

The NAV history contains 46,000 observations across 40 funds. The selected-fund trend below provides a representative view of the different paths observed in the project dataset.

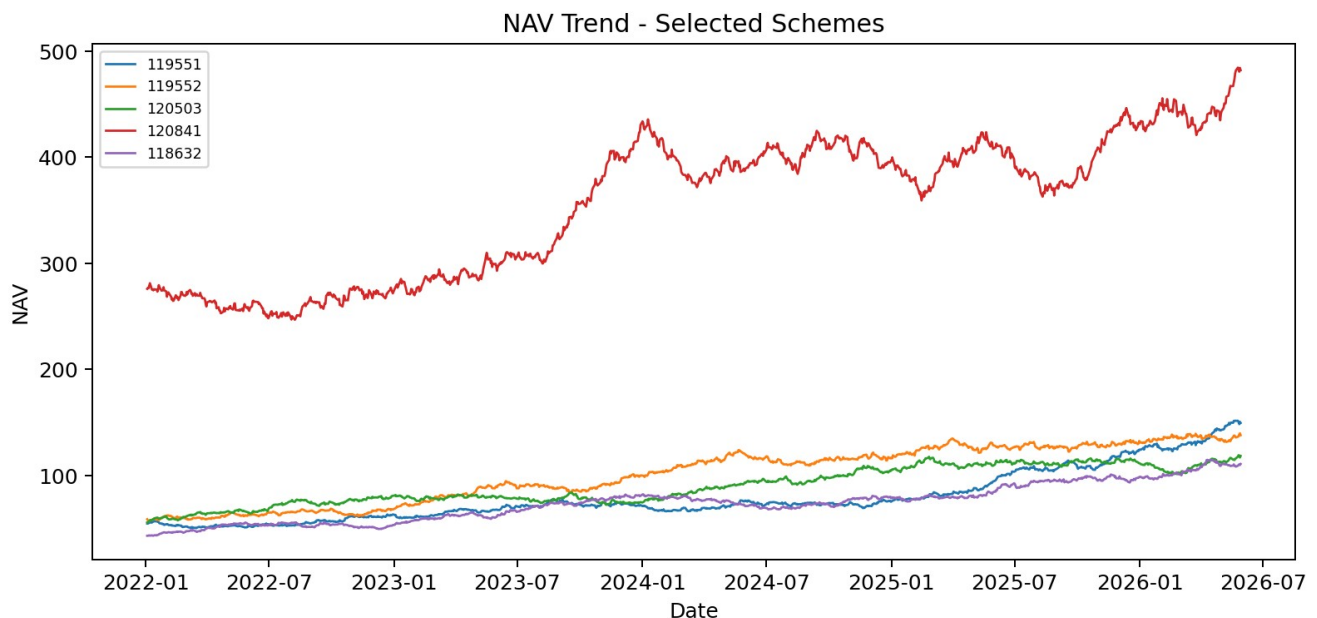


Figure 2. NAV trend for five selected schemes from the project NAV history.

8.2 AUM by Fund House

AUM is an aggregated fund-house measure. The chart below ranks the latest available snapshot from the supplied AUM dataset and helps identify relative scale across fund houses.

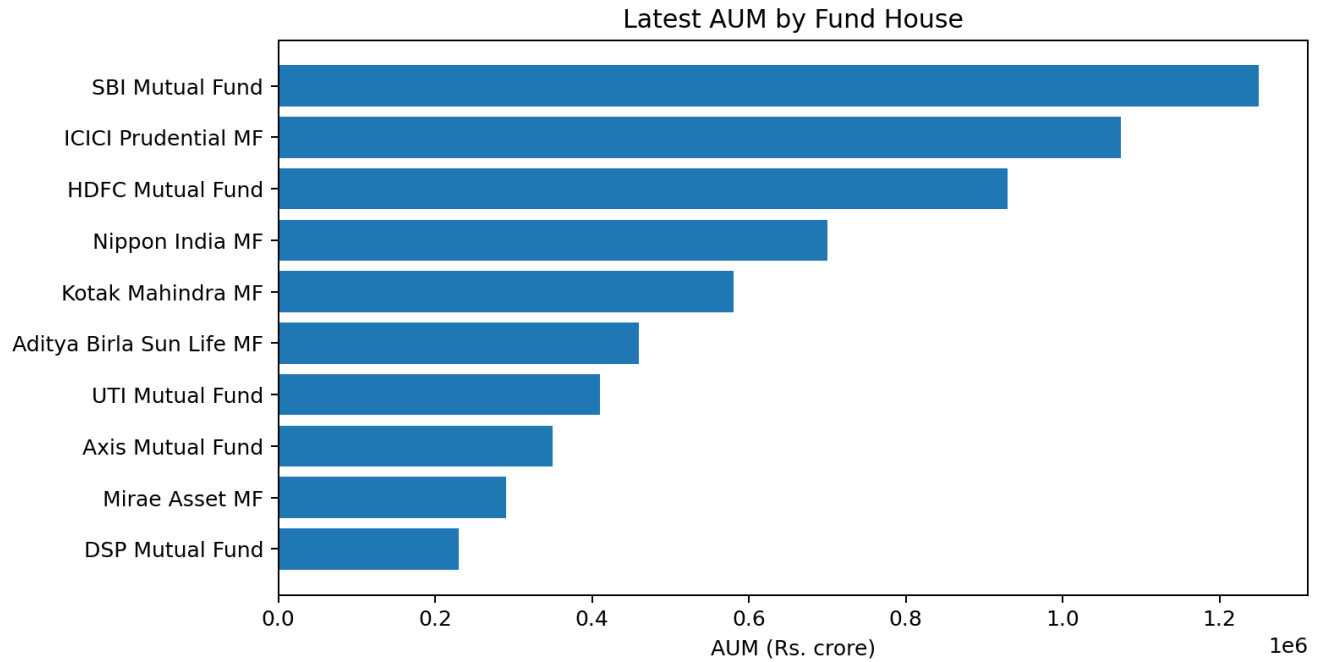


Figure 3. Latest available AUM by fund house.

8.3 SIP and Folio Trends

SIP inflow measures the flow of systematic investment into mutual funds, while folio count reflects the number of investor accounts. They answer different business questions and are intentionally analysed separately.

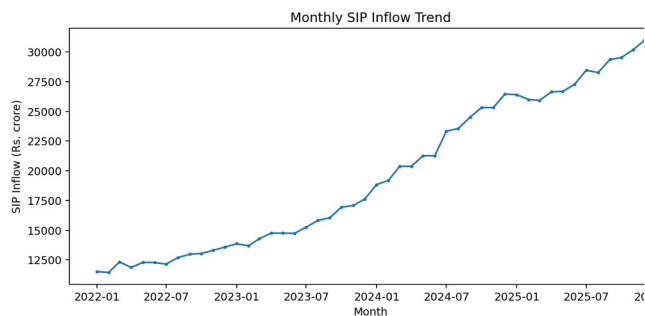


Figure 4. Monthly SIP inflow trend.

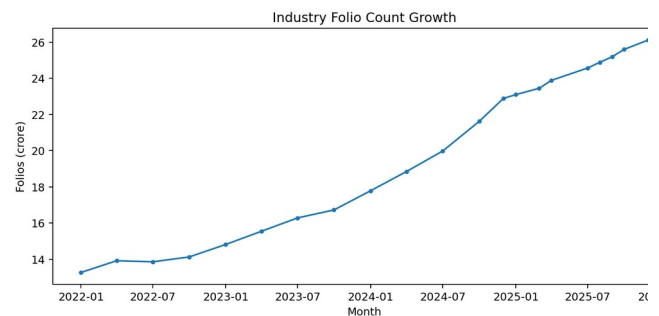


Figure 5. Industry folio-count growth.

8.4 Return Distribution

The observed daily-return series contains 45,960 observations. The mean daily return is 0.0006311 with standard deviation 0.0102904; the empirical range is -0.0581 to 0.0647.

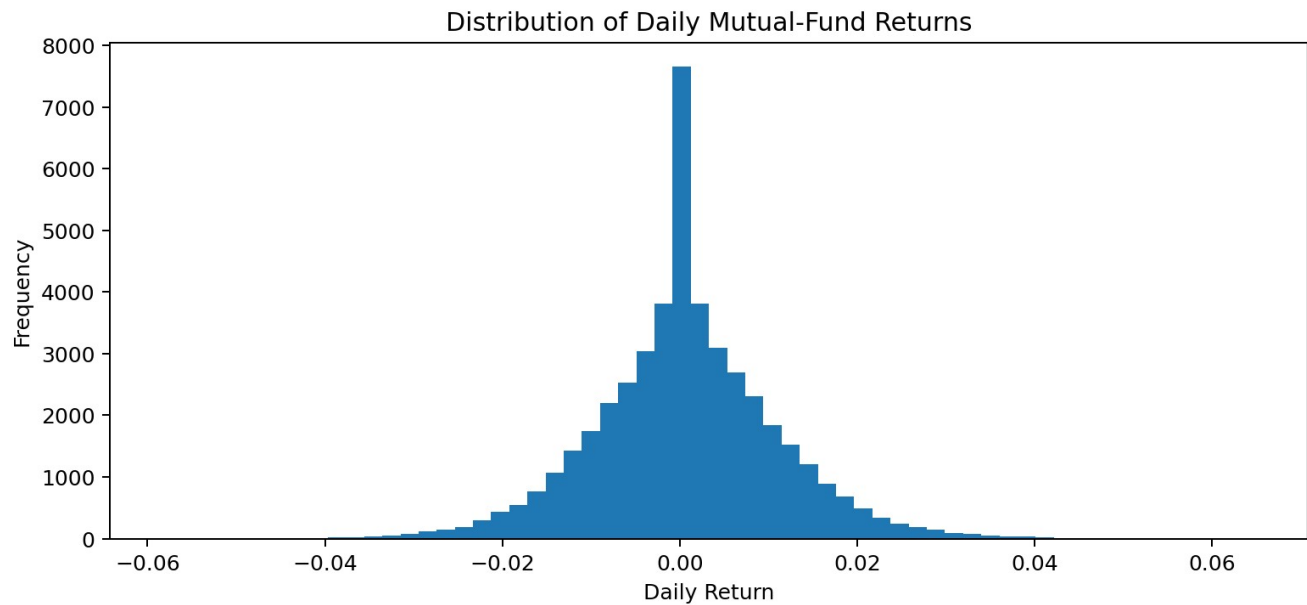


Figure 6. Distribution of daily NAV returns.

8.5 Investor and Geographic Analysis

The investor transaction data contains age group, gender, state, city, city tier, annual income, payment mode and KYC status. The dashboard and notebook analyses use these fields to examine how transaction behaviour differs across segments.

For SIP geography, the analysis aggregates SIP transaction amount by state and separates T30 and B30 city tiers. The purpose is descriptive segmentation rather than causal inference.

8.6 Category Inflow Analysis

Category inflows reveal where investor money is moving across Large Cap, Mid Cap, Small Cap, ELSS, Liquid, Flexi Cap and other categories. The Power BI dashboard includes a category inflow heatmap and a top-five category view.

8.7 Correlation and Sector Concentration

Pairwise correlation of daily returns identifies how funds move together. Correlation close to +1 means the funds tend to move together; lower correlation can improve diversification when combined with appropriate risk and return characteristics. Sector allocation and HHI then show whether a fund's portfolio is concentrated in a small number of sectors.

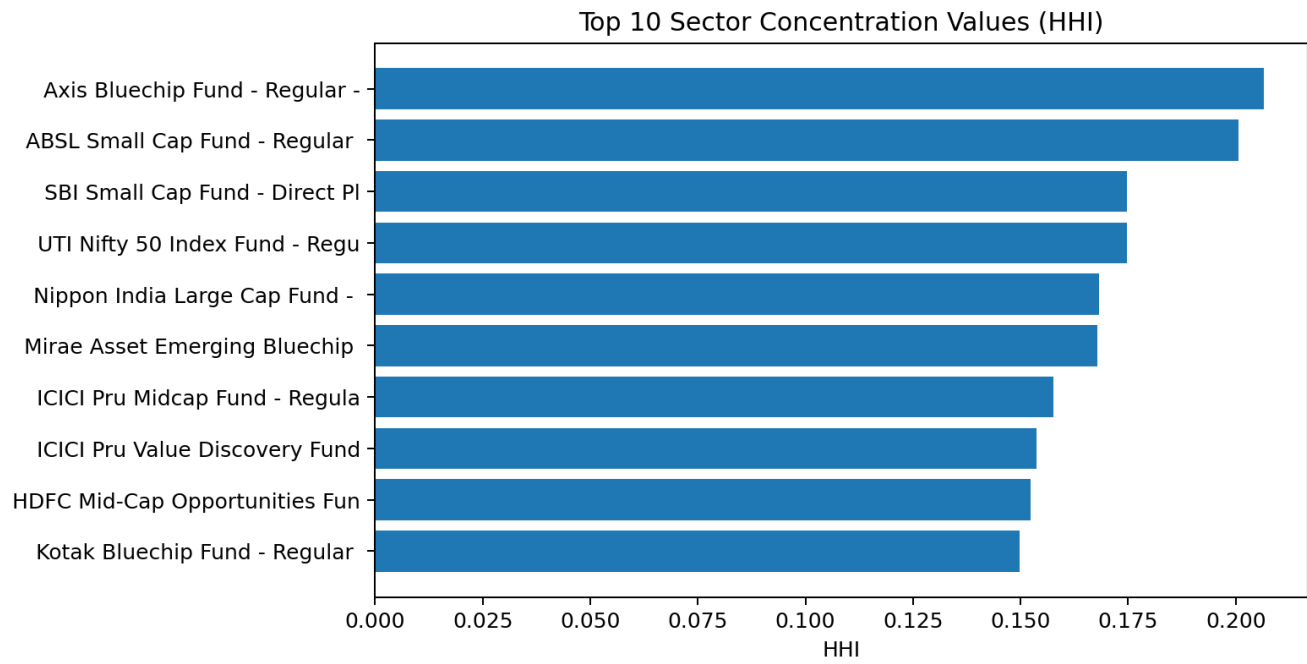


Figure 7. Highest sector concentration values using HHI from the portfolio holdings dataset.

9. Fund Performance and Risk Analytics

The performance layer applies financial formulas to the daily NAV series and the supplied benchmark series. The objective is to produce comparable measures of return, risk, benchmark-relative performance and downside.

9.1 Daily Return Validation

The NAV dataset contains 46,000 rows and produces 45,960 non-null daily-return observations after the first observation of each scheme is removed. There are no missing NAV values, duplicate NAV rows or duplicate fund-date records in the loaded dataset.

Statistic	Value
N	45,960
Min	-0.0581020
Max	0.0647131
Mean	0.0006311
Median	0.0003402
Std	0.0102904

9.2 CAGR

CAGR is used to compare growth across a common annualised scale. The project also stores 1-year, 3-year and 5-year returns in the scheme-performance dataset.

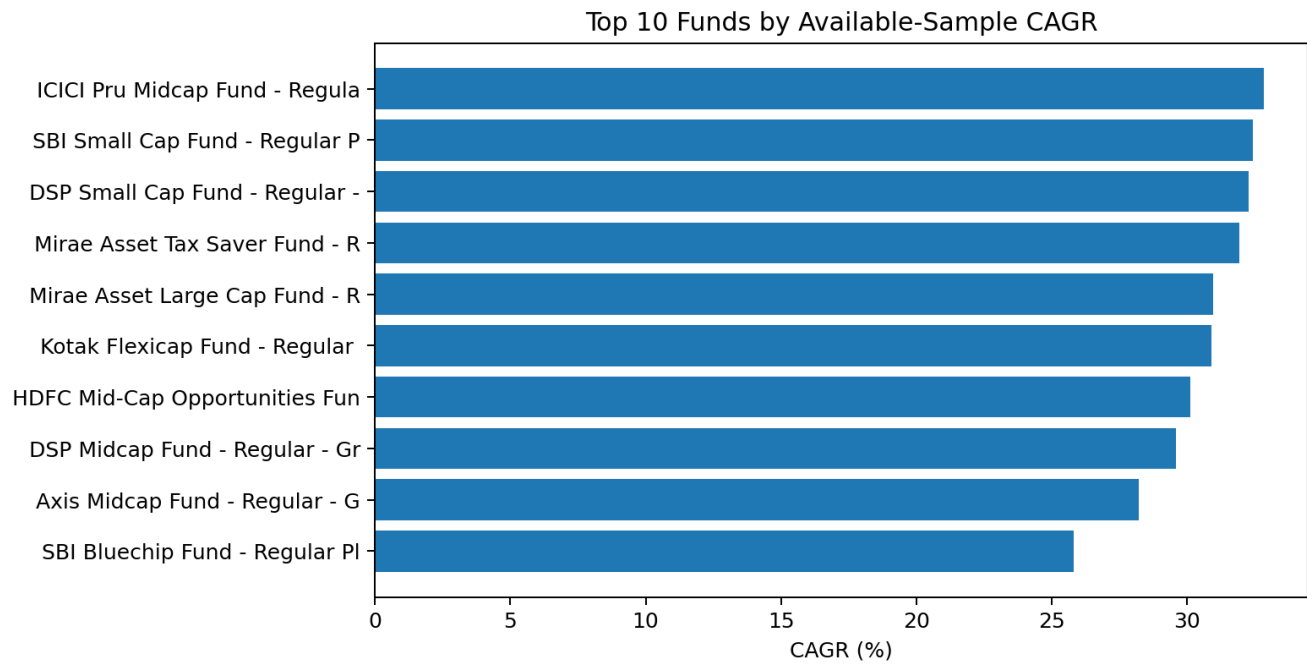


Figure 8. Ten highest available-sample CAGR values calculated from NAV start/end observations.

9.3 Sharpe and Sortino

Sharpe treats total standard deviation as the risk denominator. Sortino focuses on negative-return dispersion. Together they make it possible to compare funds that may have similar headline returns but materially different risk behaviour.

9.4 Alpha, Beta and Drawdown

Alpha and Beta are based on regression against NIFTY 100 returns. Maximum Drawdown identifies the worst historical peak-to-trough loss. The performance dataset also stores annualised standard deviation, benchmark return and expense ratio, enabling a composite scorecard.

9.5 Fund Scorecard

The capstone specifies a 0-100 composite score: 30% 3-year return rank + 25% Sharpe rank + 20% Alpha rank + 15% inverse expense-ratio rank + 10% inverse maximum-drawdown rank. The weighting reflects a balance between growth, risk-adjusted quality, excess return, cost and downside protection.

9.6 Benchmark and Tracking Error

The project compares selected fund paths with NIFTY 50 and NIFTY 100 and computes tracking error. The execution screenshots supplied with the project show both the normalized benchmark comparison and the tracking-error table for five funds.

10. Advanced Analytics and Investor Behaviour

10.1 Historical VaR and CVaR

Historical VaR and CVaR extend the risk analysis beyond standard deviation. The empirical tail results identify the schemes with the most negative 5th-percentile returns and the average loss in the worst 5% tail.

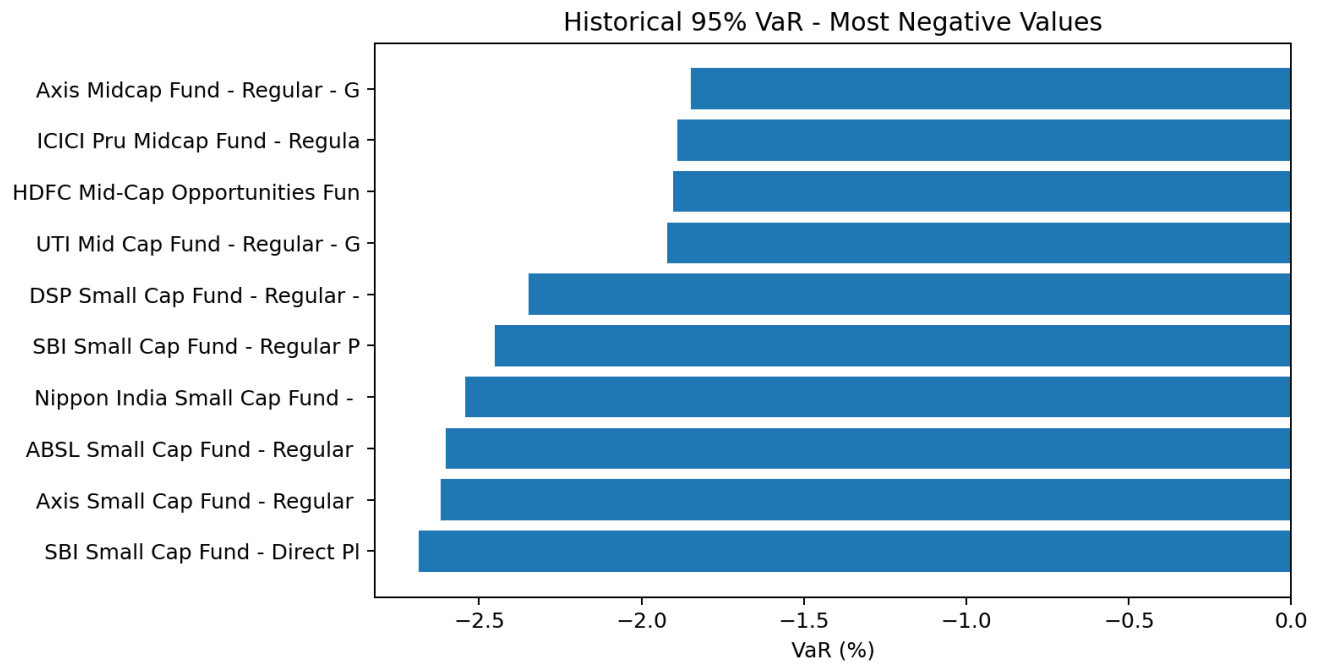


Figure 9. Historical 95% VaR - ten most negative values in the supplied return distributions.

10.2 Rolling 90-Day Sharpe

The rolling Sharpe view evaluates the change in risk-adjusted performance through time. The supplied execution proof shows the project calculating 5,750 rolling-Sharpe observations for five selected schemes.

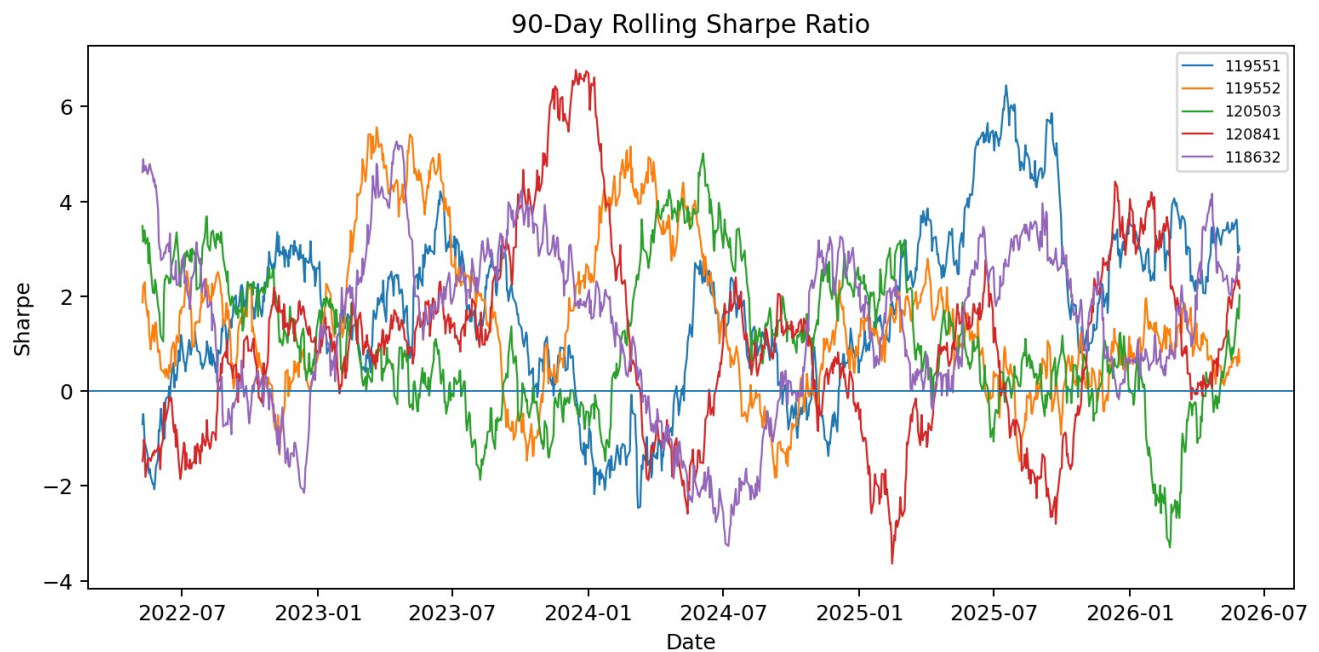


Figure 10. 90-day rolling Sharpe ratio for five selected funds.

10.3 Investor Cohorts

Cohort analysis groups investors by the year of their first transaction. The project screenshots show 2024 and 2025 cohorts and compare total invested amount, average SIP amount and top-fund preference.

Cohort	Investors	Transactions	Total SIP amount (Rs.)	Avg. SIP (Rs.)
--------	-----------	--------------	------------------------	----------------

2024	4456	19313	212,189,262	10,986.86
2025	306	403	5,044,229	12,516.70

10.4 SIP Continuity

There are 19,716 SIP transactions across 4,762 unique SIP investors. 1,362 investors have at least six SIP transactions and therefore qualify for continuity analysis. Under the project rule, 30 are regular and 1,332 are at-risk, resulting in a 97.80% at-risk rate.

Because the transaction dataset is synthetic, this percentage is a demonstration of the analytical method and should not be interpreted as a real industry retention estimate.

10.5 Recommender Logic

The simple recommender accepts a Low, Moderate or High risk appetite, filters the fund universe to the corresponding risk grade, and selects the top three funds by Sharpe ratio. The rule is transparent and easy to audit but is not a personalised financial-advice system.

10.6 Sector HHI

Higher HHI values indicate greater concentration in a small set of sectors. The metric is used as a portfolio-risk diagnostic alongside return and drawdown measures.

11. Power BI Dashboard

The Power BI layer turns the analytical outputs into an executive-facing interface. The capstone requires four pages: Industry Overview, Fund Performance, Investor Analytics, and SIP & Market Trends, with slicers, drill-through and tooltips. The supplied project screenshots show all four rendered pages.

11.1 Page 1 - Fund Performance / Scorecard

The supplied performance page presents KPI cards for total schemes, average expense ratio, average AUM, fund houses and average five-year return. It also includes fund-house, plan and category slicers, a return-versus-risk scatter, category-return comparison, risk-grade distribution, AUM by fund house, NAV history and a sortable scorecard table.

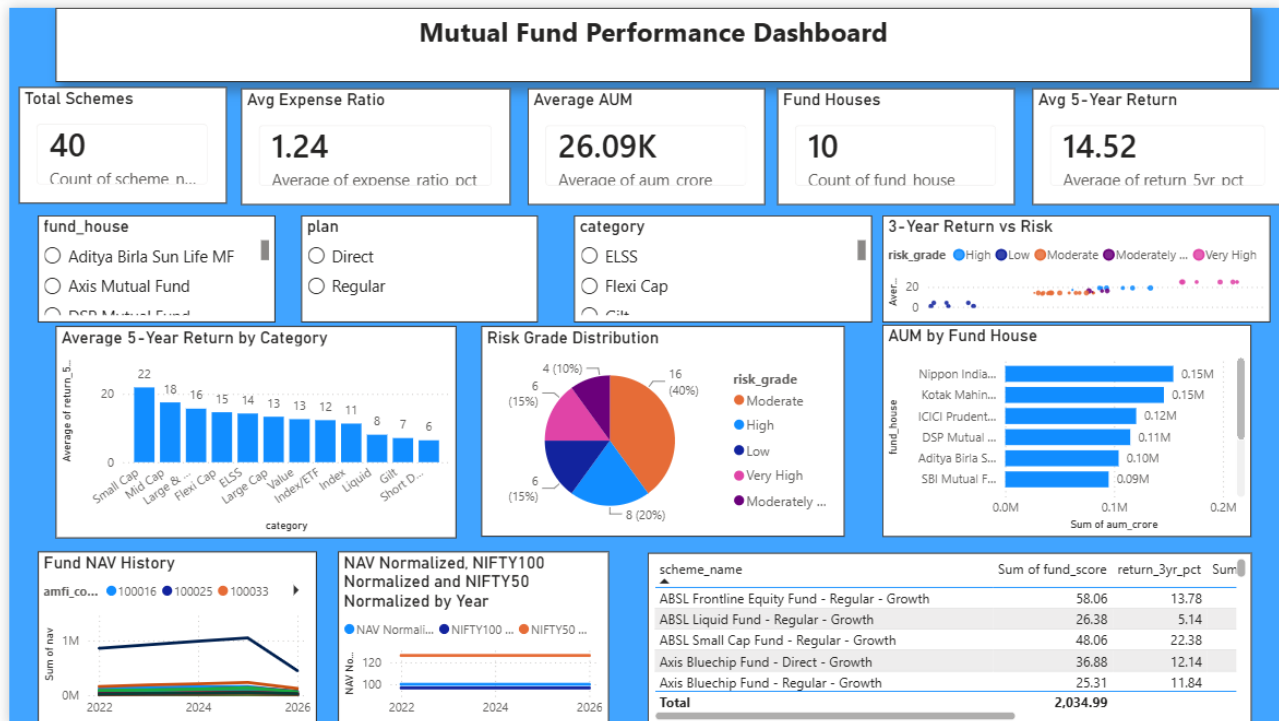


Figure 11. Power BI Page - Fund Performance dashboard.

11.2 Page 2 - Industry Overview

The industry page combines folio count, scheme count, total AUM, SIP inflows, industry AUM trend and AUM by fund house. The screenshot shows the dashboard values for the current loaded model; these should be interpreted as PBIX aggregations rather than direct copies of the industry headline values in the brief.



Figure 12. Power BI Page - Industry Overview.

11.3 Page 3 - Investor Analytics

The investor page provides state, age-group and city-tier slicers and visualises monthly transaction volume, transaction-type distribution and transaction amount by state. This page operationalises the investor-behaviour objective.

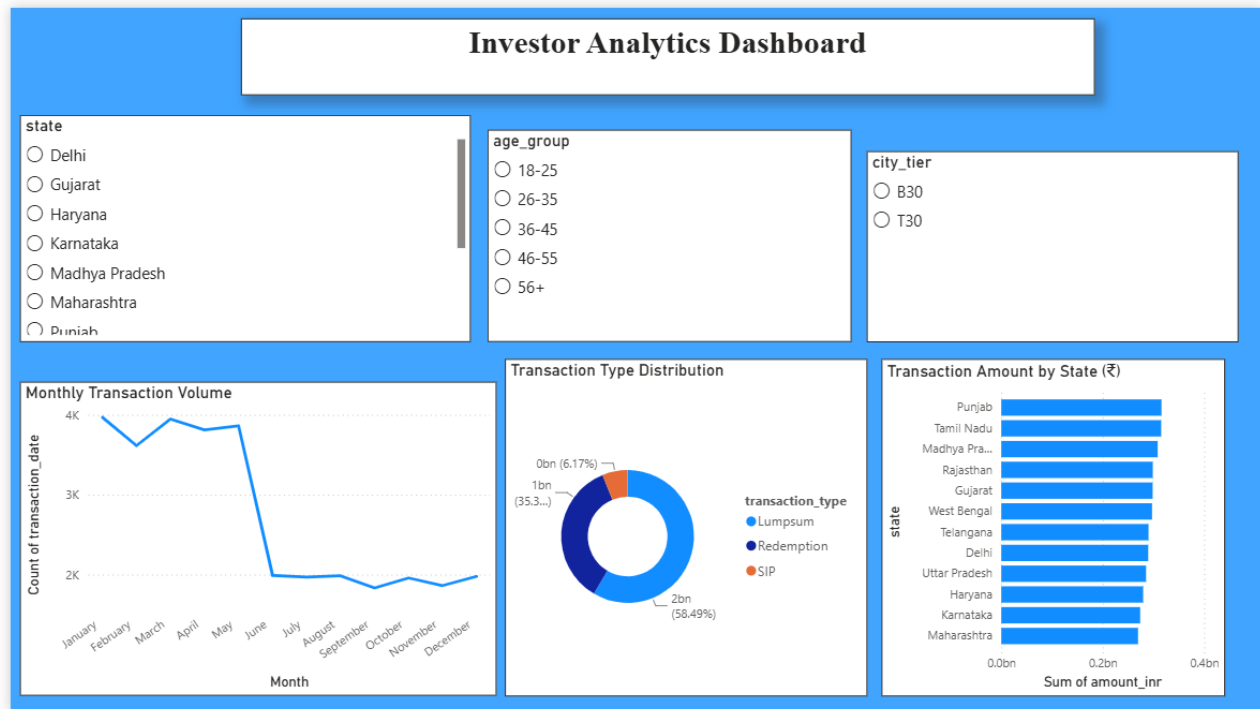


Figure 13. Power BI Page - Investor Analytics.

11.4 Page 4 - SIP & Market Trends

The SIP and market page combines SIP inflow with NIFTY 50, ranks the top five categories by net inflow, shows a monthly SIP trend, and provides a category inflow heatmap/table. This page connects industry cash flows with market context.

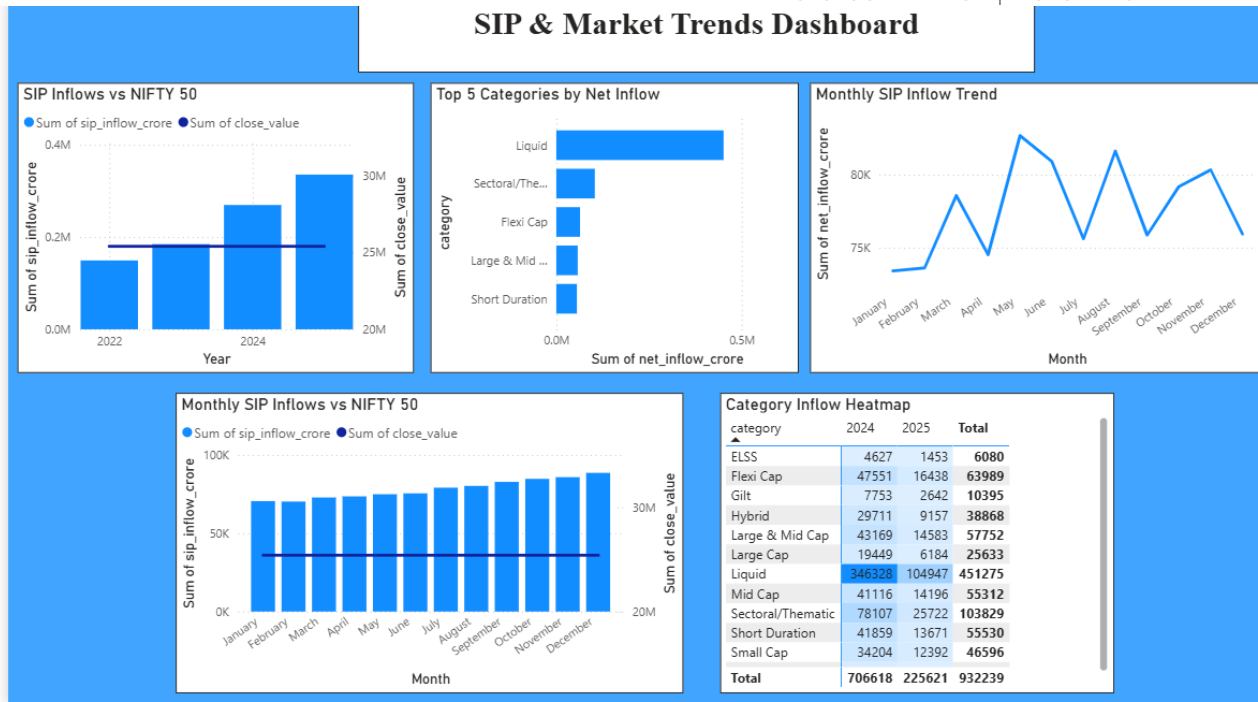


Figure 14. Power BI Page - SIP & Market Trends.

12. Findings and Business Recommendations

Finding	Interpretation	Recommendation
Finding 1	The NAV dataset is complete for the loaded sample, with 46,000 observations, zero missing values and zero duplicate fund-date records.	Retain automated data-quality checks as a mandatory ETL gate.
Finding 2	Daily returns are tightly centred around zero but include both adverse and positive tail observations.	Use both volatility and tail-risk measures rather than relying on average return.
Finding 3	CAGR, Sharpe, Sortino and Alpha answer different questions and can lead to different rankings.	Keep a composite scorecard with transparent weights.
Finding 4	Benchmark-relative analysis is necessary because absolute return alone does not reveal whether a fund added value versus its reference index.	Keep benchmark comparison and tracking error on the Fund Performance page.
Finding 5	Historical VaR/CVaR identifies funds with more negative downside-tail exposure.	Use VaR/CVaR as a monitoring layer alongside maximum drawdown.
Finding 6	Rolling Sharpe varies through time across selected funds.	Use rolling metrics to detect changing risk-adjusted conditions.
Finding 7	Investor cohorts provide a structured way to compare newer and earlier investor behaviour.	Use cohort reporting for engagement and product-preference monitoring.
Finding 8	Under the project continuity rule, a high share of eligible synthetic investors are classified as at-risk.	Treat this only as a demonstration flag and validate against real customer data before operational use.
Finding 9	Sector HHI provides a simple concentration diagnostic.	Combine HHI with sector limits or diversification alerts where appropriate.
Finding 10	The four-page dashboard unifies the analysis for executive consumption.	Preserve slicers, drill-through and tooltips and refresh from the ETL output.

13. Limitations and Data Authenticity

- The investor transaction dataset is simulated for educational purposes and should not be interpreted as real investor behaviour.

- Historical performance and risk measures are backward-looking and are not guarantees of future returns.
- The project uses a 6.5% risk-free-rate proxy for Sharpe as required by the brief; changing the proxy changes the metric.
- The 35-day SIP continuity threshold is a project-defined rule, not a validated retention model.
- Historical VaR/CVaR are distribution-based and depend on the sample period and data quality.
- Power BI totals can differ from the public industry headline values because dashboard KPIs depend on the loaded tables, filters and aggregation logic.

14. Capstone Deliverables and Evidence Matrix

Deliverable	Weight	Requirement	Evidence in submission
D1 ETL Pipeline	15%	Implemented project scripts and data ingestion workflow	Project structure / ingestion notebook evidence
D2 SQLite Database	10%	SQLite database + schema + SQL queries	bluestock_mf.db, schema.sql, queries.sql visible in project
D3 EDA Notebook	15%	EDA charts + findings	EDA notebook and generated chart files
D4 Performance Metrics	15%	Daily returns, CAGR, Sharpe, Sortino, Alpha/Beta, drawdown, scorecard, benchmark	Performance_Analytics notebook + execution screenshots
D5 Interactive Dashboard	20%	4 pages + slicers + drill-through + tooltips	4 rendered Power BI screenshots in Appendix A
D6 Advanced Analytics	10%	VaR/CVaR, rolling Sharpe, cohorts, SIP continuity, recommender, HHI	Advanced_Analytics notebook + execution proof
D7 Report + Slides	15%	Professional report and 12-slide presentation	This report + presentation

15. Conclusion and Future Scope

The project demonstrates a complete mutual-fund analytics workflow from ingestion through business presentation. The main strength is the integration of multiple perspectives: time-series performance, risk-adjusted return, benchmark comparison, downside-tail risk, investor segmentation, SIP continuity and sector concentration. This is more useful for decision support than a return-only report.

The next practical improvements are to fully automate the ETL on a schedule, add a Streamlit alternative interface, build Monte Carlo projections, implement Markowitz portfolio optimisation, automate weekly HTML email reporting and strengthen investor-retention models with validated real-world customer data.

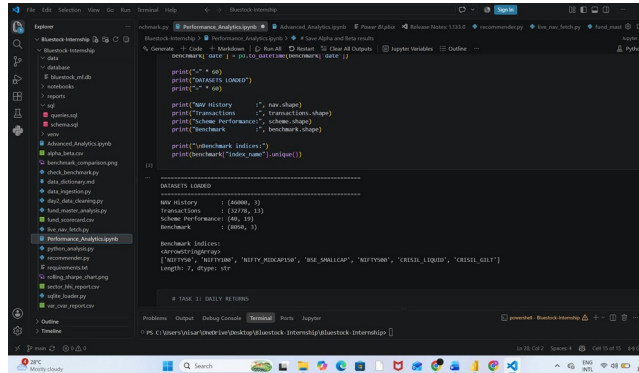
Overall, the solution meets the intended architecture of Extract -> Transform -> Load -> Analyse -> Visualise and produces a cohesive analytical foundation for executive-level mutual-fund decision support.

References

- Bluestock Fintech, Mutual Fund Analytics Capstone Project brief, uploaded project document, sections covering project objectives, datasets, ETL architecture, 7-day tasks, dashboard, advanced analytics and final report requirements.
- AMFI India - mutual-fund industry, AUM, SIP and folio information (as referenced in the capstone brief).
- mfapi.in - historical NAV API (as referenced in the capstone brief).
- mfddata.in - NAV and expense-ratio API (as referenced in the capstone brief).
- NSE India and BSE India - benchmark index data sources (as referenced in the capstone brief).

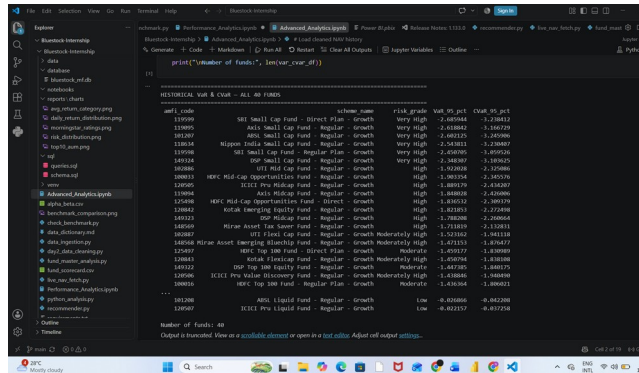
Appendix A - Selected Execution Proof

The report body contains the detailed analytical theory, generated charts and all four Power BI pages. This appendix provides a compact evidence-at-a-glance view of the most important notebook executions.



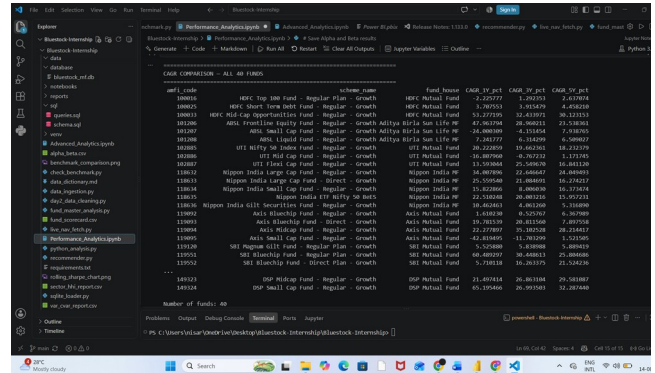
A1. Dataset loading

46,000 NAV rows, 32,778 transactions, 40 scheme-performance rows and benchmark data are shown as loaded.



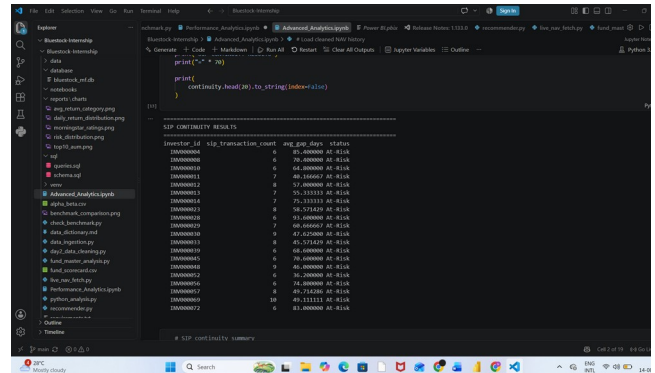
A3. VaR / CVaR execution

Historical 95% VaR and CVaR values are shown for the fund universe.



A2. CAGR execution

The notebook output shows the 40-fund CAGR comparison with 1Y/3Y/5Y fields.



A4. SIP continuity execution

The notebook shows 1,362 eligible investors, 30 regular investors, 1,332 at-risk investors and a 97.80% project-defined at-risk rate.

Evidence interpretation: screenshots demonstrate that the notebooks executed and produced the displayed outputs. Financial correctness still depends on the formulas, assumptions and source data documented in the main report.